



NUST CHIP DESIGN CENTRE

RISC-V Architecture

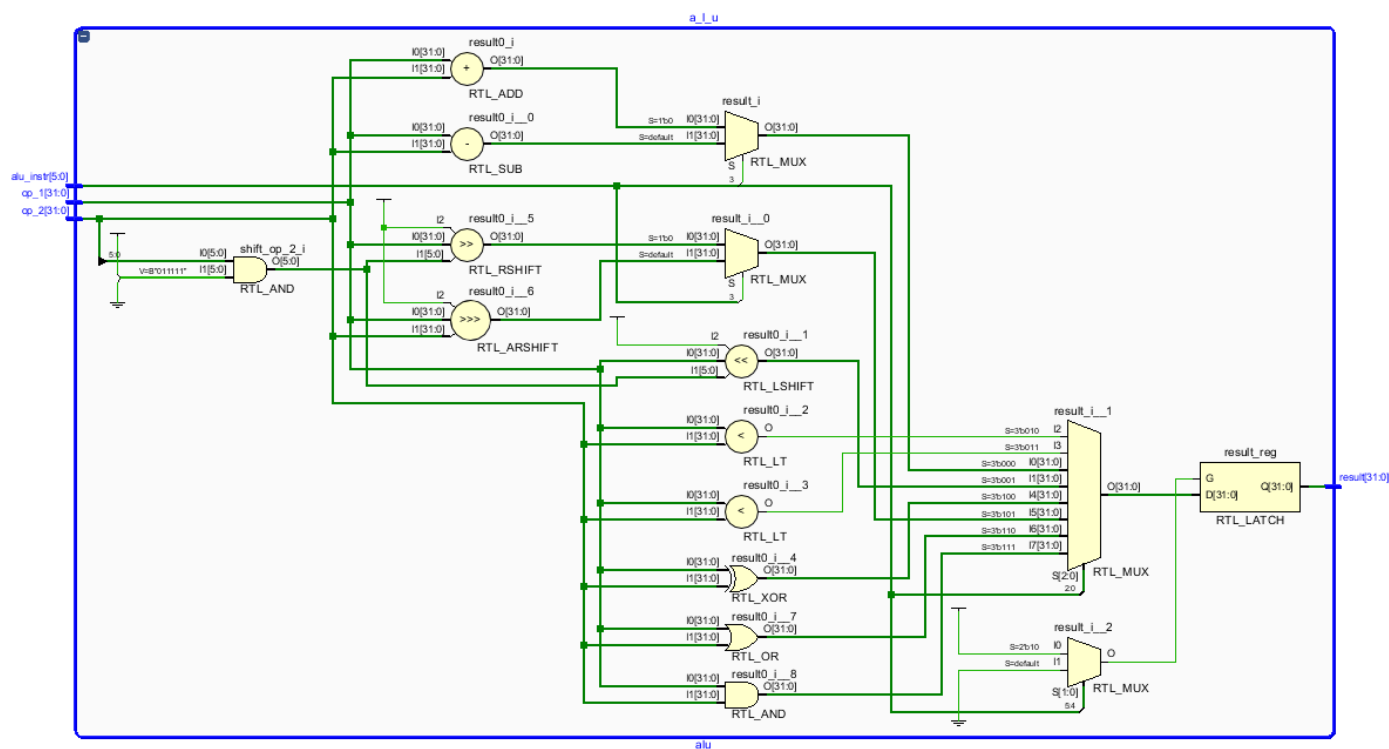
Lab Manual # 18

RISC-V Datapath Implementation **(I-Type and R-Type Instructions)**

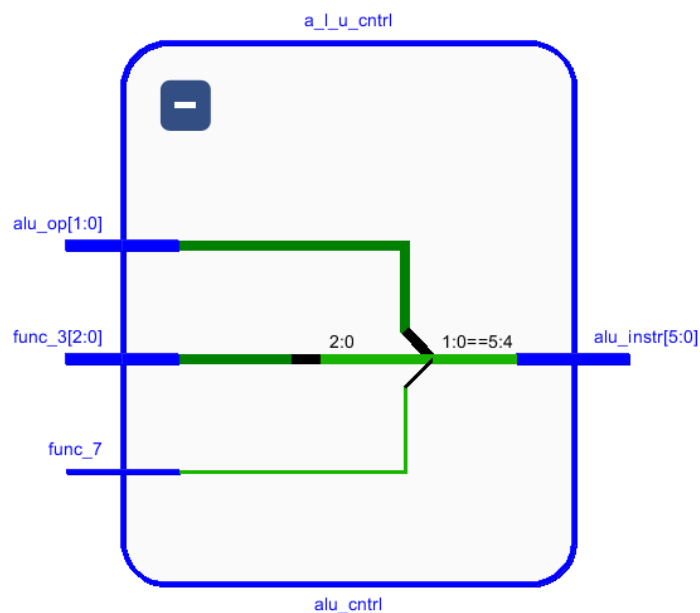
<u>Name</u>	<i><u>Muddassir Ali Siddiqui</u></i>
<u>Instructor</u>	<i><u>Sir Imran & Sir Umer</u></i>
<u>Date</u>	<i><u>27th Aug 2025</u></i>

1. In-Lab Tasks: (Write your lab task & screenshots here)

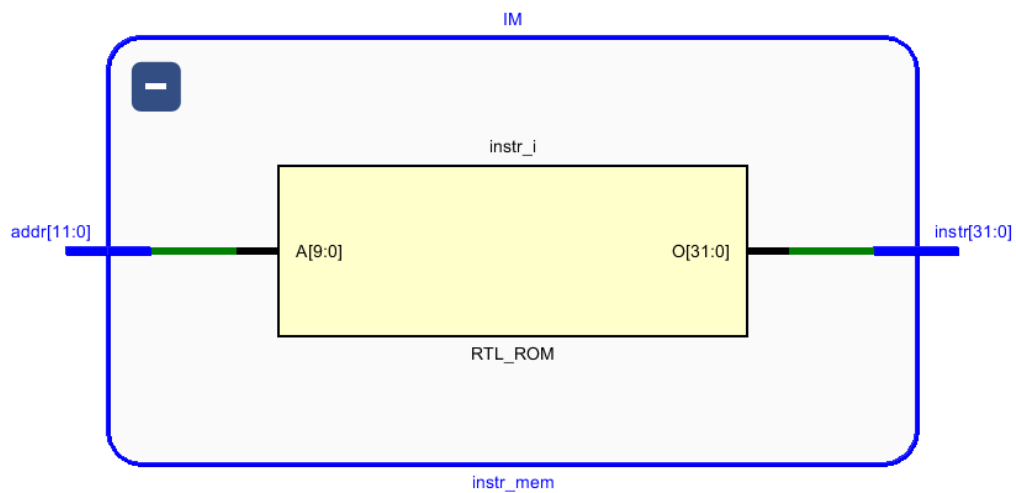
i. ALU



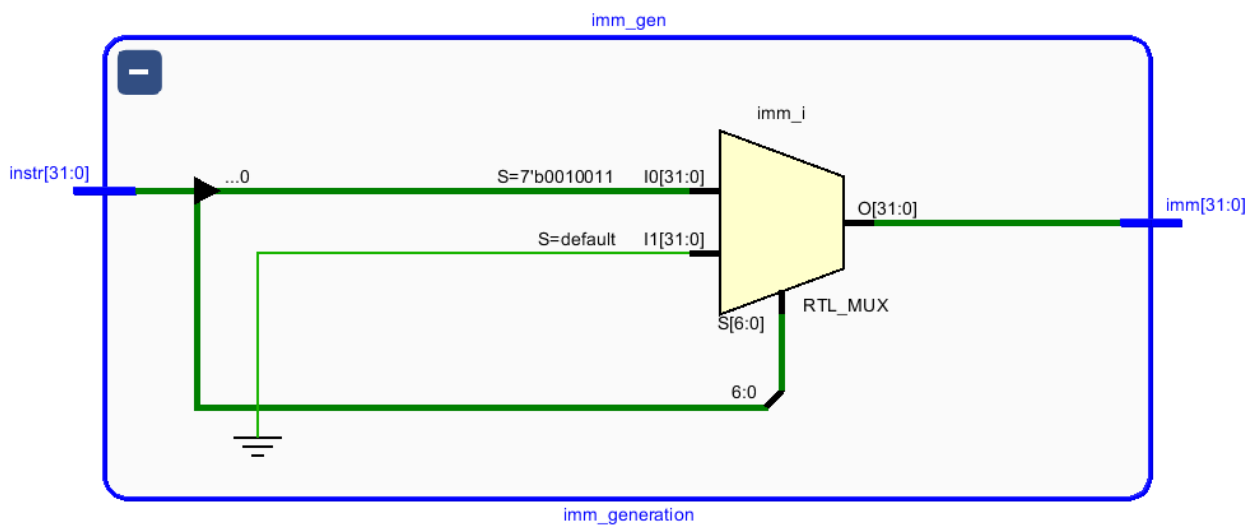
ii. ALU Control:



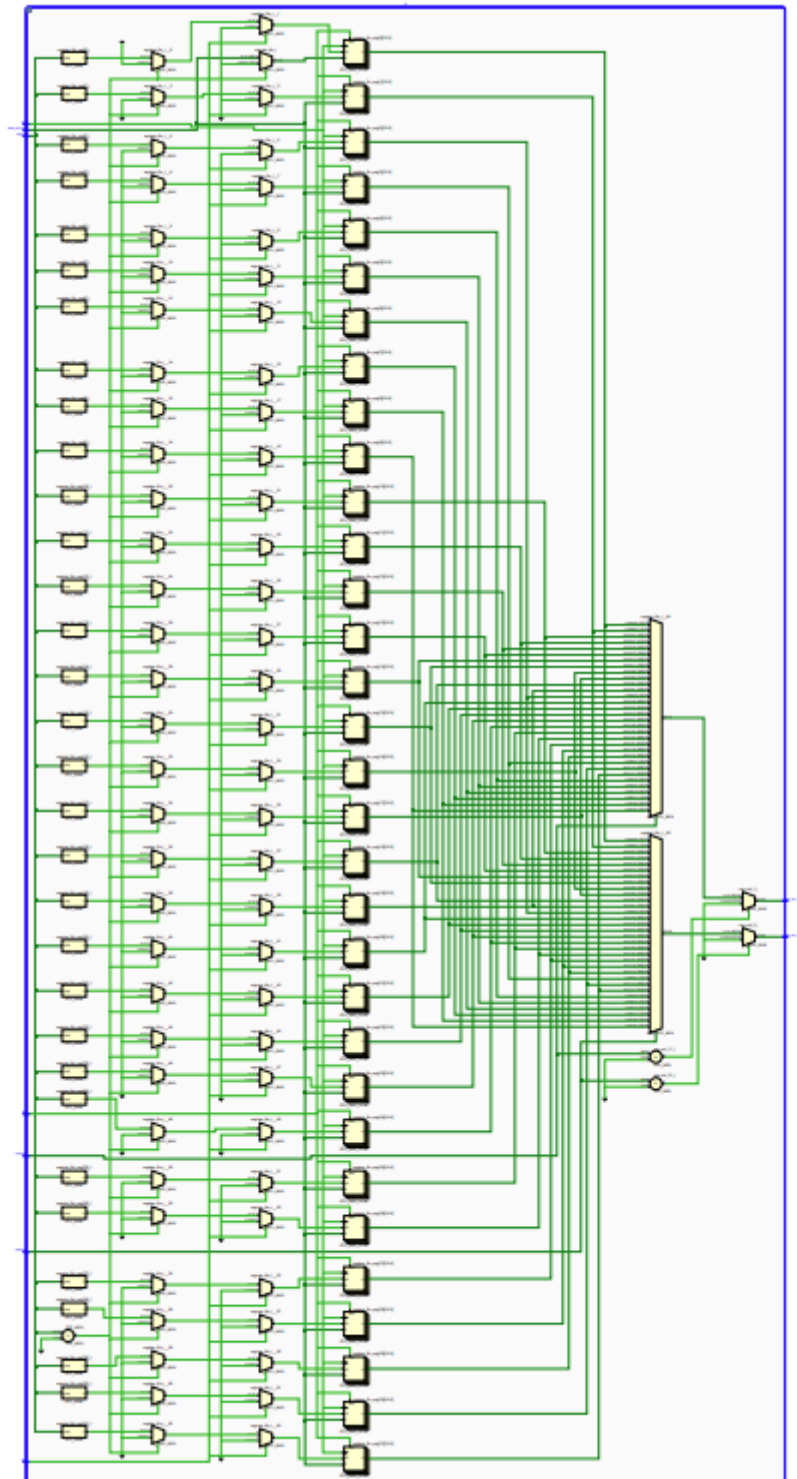
iii. Instruction Memory:



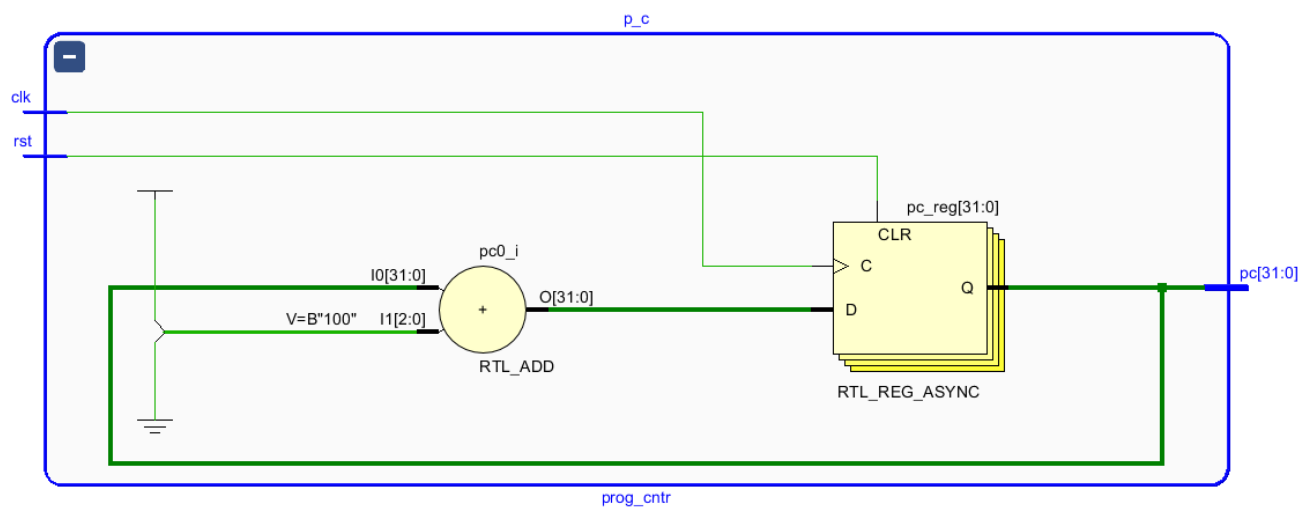
iv. Immediate Generation:



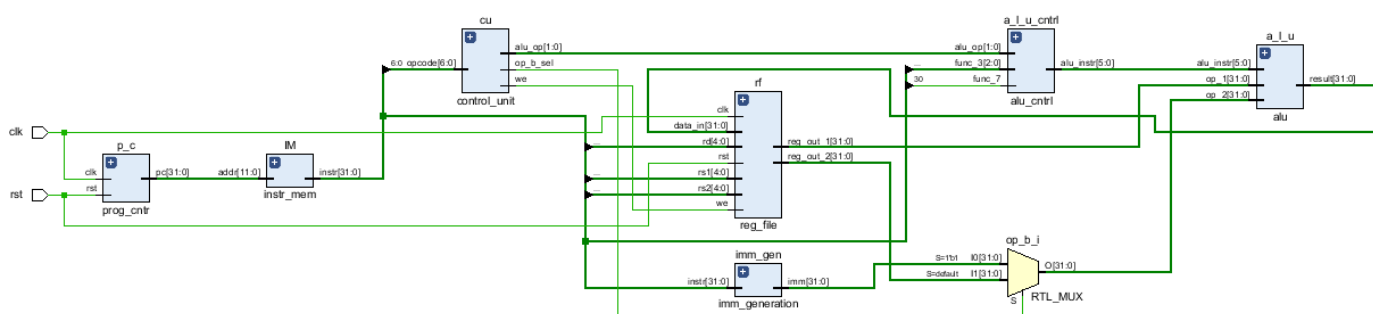
v. Register File:



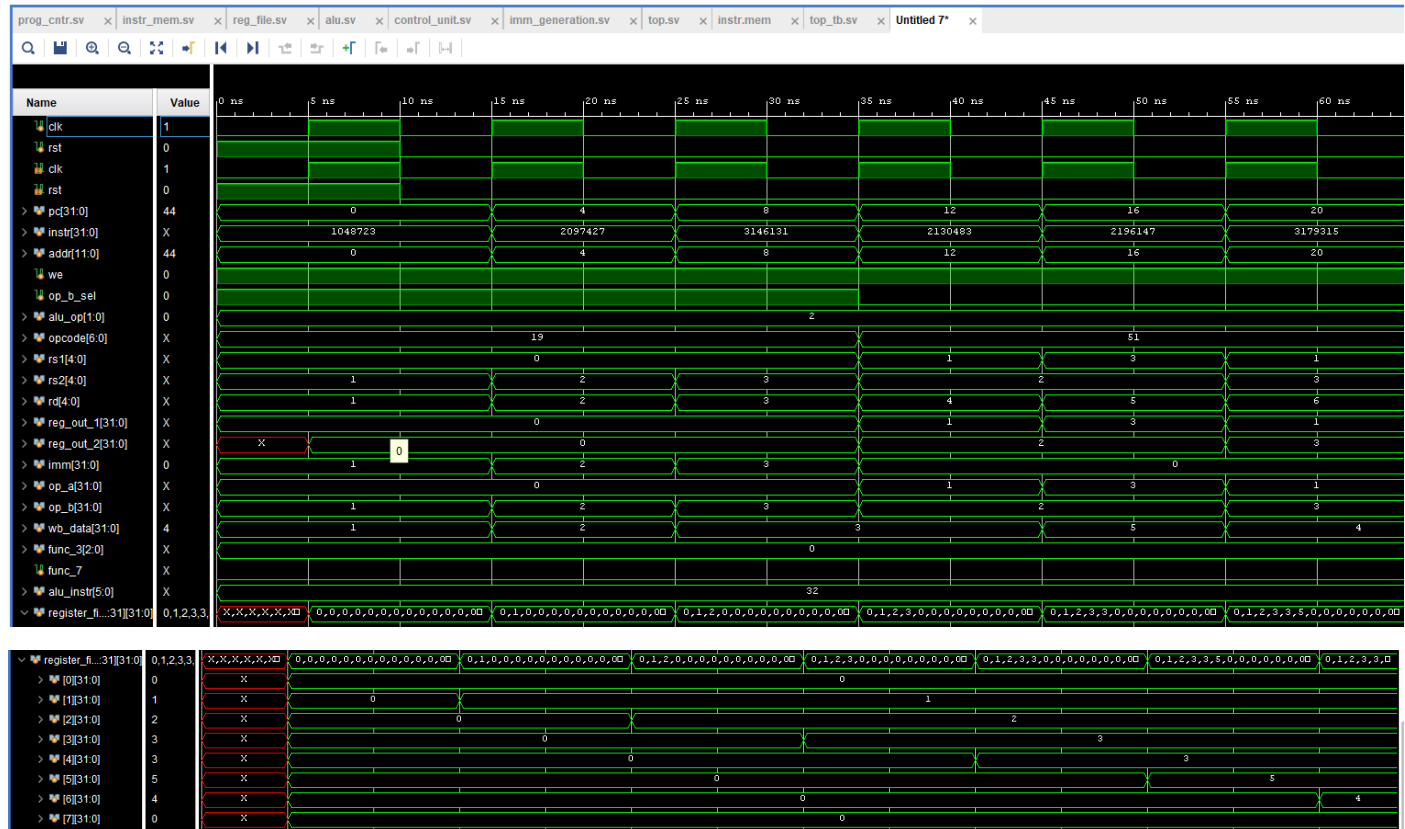
vi. Program Counter:



vii. Top:



viii. Waveforms



2. Critical Analysis: (Write you critical analysis / conclusion here)

In the current stage of development, the RV32I single-cycle processor core has been partially implemented. The instruction memory module has been designed to fetch 32-bit instructions from an external hex file, and the control unit has been developed to decode the fetched instruction. At this point, the design is limited to handling **R-type and I-type instructions**, allowing the control signals for arithmetic, logical, and immediate-based operations to be generated. A basic fetch-decode path is functional, confirming that the instruction flow through the memory and control unit is established.